

§ 27. Multiplication of determinants.

The theorem which we still have to prove teaches how one can represent the product of two determinants with the same number of rows by a single determinant with the same number of rows. One is led to it most simply by considering the solution of two systems of linear equations.

Let the coefficients $a_i^{(k)}, b_h^{(k)}$, when i, h run through the sequence of numbers $1, 2, \dots, n$, be arbitrary variable quantities; likewise let the quantities x_i, y_i, z_i be bound only by the relations

$$\sum_i a_i^{(k)} x_i = y_k, \quad (1)$$

$$\sum_k b_h^{(k)} y_k = z_h. \quad (2)$$

If the problem is now posed of determining the variables x by the variables z , this problem can be solved in two ways. By § 26, (4) one may solve equations (1) with respect to x , equations (2) with respect to y , and insert the latter expressions into the former. If A, B denote the determinants of the two systems of equations, that is,

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \cdots & \cdots & \cdots & \cdots \\ a_1^{(n)} & a_2^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \quad B = \begin{vmatrix} b_1^{(1)} & b_2^{(1)} & \cdots & b_n^{(1)} \\ b_1^{(2)} & b_2^{(2)} & \cdots & b_n^{(2)} \\ \cdots & \cdots & \cdots & \cdots \\ b_1^{(n)} & b_2^{(n)} & \cdots & b_n^{(n)} \end{vmatrix},$$

and if $A_i^{(k)}, B_i^{(k)}$ denote the first subdeterminants, one obtains

$$Ax_k = \sum_i A_k^{(i)} y_i, \quad (3)$$

$$By_i = \sum_h B_i^{(h)} z_h, \quad (4)$$

and, by substitution of (4) in (3),

$$ABx_k = \sum_h z_h \sum_i A_k^{(i)} B_i^{(h)}. \quad (5)$$

But one may also proceed by inserting from (1) the expressions for y into (2), whereby, if

$$c_i^{(h)} = \sum_s a_i^{(s)} b_h^{(s)} \quad (6)$$

is put, one obtains

$$\sum_i c_i^{(h)} x_i = z_h. \quad (7)$$

Now put

$$C = \begin{vmatrix} c_1^{(1)} & c_2^{(1)} & \cdots & c_n^{(1)} \\ c_1^{(2)} & c_2^{(2)} & \cdots & c_n^{(2)} \\ \cdots & \cdots & \cdots & \cdots \\ c_1^{(n)} & c_2^{(n)} & \cdots & c_n^{(n)} \end{vmatrix}. \quad (8)$$

If $C_i^{(k)}$ denotes the subdeterminants of C , the solution of (7) gives

$$Cx_k = \sum_h z_h C_k^{(h)}, \quad (9)$$

and comparison of (5) with (9) gives

$$AB = C, \quad (10)$$

and for the subdeterminants

$$C_k^{(h)} = \sum_i A_k^{(i)} B_i^{(h)}. \quad (11)$$

In this conclusion, however, there is still a gap, caused by the doubt whether (5) and (9) might not differ by a factor common to both sides. To remove this doubt we shall prove equation (10) directly; the correctness of (11) then follows.

We shall form the principal term of the determinant C from (6). Let $s_1, s_2, \dots, s_n$ be mutually independent summation letters which run from 1 to n . Then

$$\begin{aligned} c_1^{(1)} c_2^{(2)} \dots c_n^{(n)} &= \sum_{s_1} a_1^{(s_1)} b_1^{(s_1)} \sum_{s_2} a_2^{(s_2)} b_2^{(s_2)} \dots \sum_{s_n} a_n^{(s_n)} b_n^{(s_n)} \\ &= \sum_{s_1, s_2, \dots, s_n} a_1^{(s_1)} a_2^{(s_2)} \dots a_n^{(s_n)} b_1^{(s_1)} b_2^{(s_2)} \dots b_n^{(s_n)}. \end{aligned} \quad (12)$$

The permutation of the lower indices of the c corresponds to the permutation of the lower indices of the a , and if, with regard to the rule of signs, one therefore forms the determinant C , one obtains

$$\sum \pm c_1^{(1)} c_2^{(2)} \dots c_n^{(n)} = \sum_{s_1, s_2, \dots, s_n} b_1^{(s_1)} b_2^{(s_2)} \dots b_n^{(s_n)} \sum \pm a_1^{(s_1)} a_2^{(s_2)} \dots a_n^{(s_n)}. \quad (13)$$

Now

$$\sum \pm a_1^{(s_1)} a_2^{(s_2)} \dots a_n^{(s_n)}$$

is, by § 21, VI, always zero if the same digit occurs twice among $s_1, s_2, \dots, s_n$; hence only those terms in (13) keep a non-zero value for which $s_1, s_2, \dots, s_n$ is an arrangement of the indices $1, 2, \dots, n$. This value is $+A$ or $-A$ according as this arrangement belongs to the first or the second kind (§ 21, IV).

Therefore the right-hand side of (13) becomes

$$A \sum_{s_1, s_2, \dots, s_n} \pm b_1^{(s_1)} b_2^{(s_2)} \dots b_n^{(s_n)},$$

where the sum is extended over all permutations $s_1, s_2, \dots, s_n$. But this sum is precisely the determinant B , and hence the formula

$$C = AB$$

is proved.

The law of formation of the elements $c_i^{(h)}$ contained in formula (6) can be expressed in words as follows. To obtain the elements of the determinant C which is the product of the two determinants A, B , one multiplies the elements of each column of A by the corresponding elements of a column of B and adds the products.

By the theorems on determinants the form of C can be varied in many ways. We record the following. If two columns in A or in B are interchanged, the elements $c_i^{(h)}$ do not change, but only interchange among themselves. If, however, two rows in A or in B are interchanged, then the $c_i^{(h)}$ change because the factors in the individual products of the sum are grouped differently; but if corresponding interchanges are made simultaneously in the rows of A and of B , the $c_i^{(h)}$ remain unchanged, since only the individual terms of the sum are thereby interchanged.

By turning the rows into columns in A or in B , or in both at once, one obtains three further different ways of forming the product of two determinants in determinant form. This can also be expressed as follows. To form the product of two determinants, one may multiply the elements of the separate rows or columns of one factor by the corresponding elements of the rows or columns of the other factor, add the products, and regard these sums of products as the elements of a new determinant.

The multiplication rule can be applied to a product of two determinants with different numbers of elements by transforming the determinant with the smaller number of rows, by the theorem § 22, IX, into another with more rows.

§ 28. Determinants of subdeterminants.

We immediately make an application here of the multiplication law of determinants. Let, as before,

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \cdots & \cdots & \cdots & \cdots \\ a_1^{(n)} & a_2^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \quad (1)$$

and let

$$\begin{vmatrix} A_1^{(1)} & A_2^{(1)} & \cdots & A_n^{(1)} \\ A_1^{(2)} & A_2^{(2)} & \cdots & A_n^{(2)} \\ \cdots & \cdots & \cdots & \cdots \\ A_1^{(n)} & A_2^{(n)} & \cdots & A_n^{(n)} \end{vmatrix} \quad (2)$$

be the system of subdeterminants. If we form from (2) the determinant which we shall denote by Δ , we may apply the multiplication rule to the product $A\Delta$. By § 22, (3) and (7), this gives

$$A\Delta = \begin{vmatrix} A & 0 & \cdots & 0 \\ 0 & A & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & A \end{vmatrix} = A^n,$$

and, by division by A ,

$$\Delta = A^{n-1}. \quad (3)$$

Thus Δ is the $(n-1)$ st power of A . In this derivation it has indeed first been assumed that A is different from zero. But since (3) is an identity with respect to the elements $a_i^{(h)}$, that is, since it also holds when these quantities are independent variables, it follows that the formula (3) still holds in the exceptional case; that is, if A vanishes, then Δ also vanishes.

This result is a special case of a more general theorem by which any determinant of the matrix (2) may be formed. Consider the ν -rowed subdeterminant

$$\Delta_\nu = \begin{vmatrix} A_1^{(1)} & A_2^{(1)} & \cdots & A_\nu^{(1)} \\ A_1^{(2)} & A_2^{(2)} & \cdots & A_\nu^{(2)} \\ \cdots & \cdots & \cdots & \cdots \\ A_1^{(\nu)} & A_2^{(\nu)} & \cdots & A_\nu^{(\nu)} \end{vmatrix}, \quad (4)$$

from which all other ν -rowed subdeterminants can be derived by permutation of the upper and lower indices. One may apply the multiplication rule by transforming Δ_ν , according to the concluding remark of the last paragraph, into an n -rowed determinant.

Put

$$\Delta_\nu = \begin{vmatrix} A_1^{(1)} & \cdots & A_\nu^{(1)} & A_{\nu+1}^{(1)} & \cdots & A_n^{(1)} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ A_1^{(\nu)} & \cdots & A_\nu^{(\nu)} & A_{\nu+1}^{(\nu)} & \cdots & A_n^{(\nu)} \\ 0 & \cdots & 0 & 1 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ 0 & \cdots & 0 & 0 & \cdots & 1 \end{vmatrix}, \quad (5)$$

where $n - \nu$ rows and columns have been adjoined, the former containing only zeros except in the diagonal terms. If one now forms the product $A\Delta_\nu$, it follows that

$$A\Delta_\nu = A^\nu \begin{vmatrix} a_{\nu+1}^{(\nu+1)} & \cdots & a_n^{(\nu+1)} \\ \cdots & \cdots & \cdots \\ a_{\nu+1}^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \quad (6)$$

and this is by Theorem IX, § 22. Dividing here by A and applying the notation of § 23, one obtains

$$\Delta_\nu = A^{\nu-1} A_{1,2,\dots,\nu}^{1,2,\dots,\nu}. \quad (7)$$

For $\nu = 2$ this gives the special result

$$A_1^{(1)} A_2^{(2)} - A_1^{(2)} A_2^{(1)} = A A_{1,2}^{1,2}. \quad (8)$$

We shall use formula (8) often later. In the notation by differential quotients it can be represented as

$$A \frac{\partial^2 A}{\partial a_1^{(1)} \partial a_2^{(2)}} = \frac{\partial A}{\partial a_1^{(1)}} \frac{\partial A}{\partial a_2^{(2)}} - \frac{\partial A}{\partial a_1^{(2)}} \frac{\partial A}{\partial a_2^{(1)}}. \quad (9)$$

It is especially important in the case where A is a symmetric determinant, so that $a_i^{(k)} = a_k^{(i)}$; then also

$$\frac{\partial A}{\partial a_1^{(2)}} = \frac{\partial A}{\partial a_2^{(1)}},$$

where in the differentiation no account is taken of the dependence $a_i^{(k)} = a_k^{(i)}$. Then formula (9) becomes

$$A \frac{\partial^2 A}{\partial a_1^{(1)} \partial a_2^{(2)}} = \frac{\partial A}{\partial a_1^{(1)}} \frac{\partial A}{\partial a_2^{(2)}} - \left(\frac{\partial A}{\partial a_1^{(2)}} \right)^2. \quad (10)$$

§ 29. Interpolation.

In § 9, under a special assumption, we already solved the problem of determining an integral rational function which assumes prescribed values for a sufficient number of prescribed values of the argument. We shall now solve this problem generally by applying determinants. First, however, we shall put forward the following observations.

In § 22, (12), the value of the determinant

$$\begin{vmatrix} \alpha_1^{n-1} & \alpha_1^{n-2} & \cdots & \alpha_1 & 1 \\ \alpha_2^{n-1} & \alpha_2^{n-2} & \cdots & \alpha_2 & 1 \\ \cdots & \cdots & & \cdots & \cdots \\ \alpha_n^{n-1} & \alpha_n^{n-2} & \cdots & \alpha_n & 1 \end{vmatrix}$$

was found to be equal to the product of differences

$$(\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3) \cdots (\alpha_1 - \alpha_n)(\alpha_2 - \alpha_3) \cdots (\alpha_2 - \alpha_n) \cdots (\alpha_{n-1} - \alpha_n). \quad (1)$$

For brevity we now denote this product of differences by

$$[\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n], \quad (2)$$

so that the quantity $[\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n]$ changes its sign when two of the elements $\alpha_1, \alpha_2, \dots, \alpha_n$ are interchanged.

We further define an integral rational function $f(x)$ of degree n in the variable x by the equation

$$f(x) = (x - \alpha_1)(x - \alpha_2) \cdots (x - \alpha_n), \quad (3)$$

so that, by § 13,

$$\begin{aligned} f'(\alpha_1) &= (\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3) \cdots (\alpha_1 - \alpha_n), \\ f'(\alpha_2) &= (\alpha_2 - \alpha_1)(\alpha_2 - \alpha_3) \cdots (\alpha_2 - \alpha_n), \\ &\vdots \\ f'(\alpha_n) &= (\alpha_n - \alpha_1)(\alpha_n - \alpha_2) \cdots (\alpha_n - \alpha_{n-1}). \end{aligned} \quad (4)$$

In the product of all these expressions each factor of the product (1) occurs twice with opposite sign; since the number of factors of (1) is $\frac{n(n-1)}{2}$, we obtain the equation

$$f'(\alpha_1)f'(\alpha_2)\cdots f'(\alpha_n) = (-1)^{\frac{n(n-1)}{2}}[\alpha_1, \alpha_2, \dots, \alpha_n]^2. \quad (5)$$

But with the aid of the relations (4) we may also represent our product (1) as follows:

$$\begin{aligned} [\alpha_1, \alpha_2, \dots, \alpha_n] &= f'(\alpha_1)[\alpha_2, \alpha_3, \dots, \alpha_n] \\ &= -f'(\alpha_2)[\alpha_1, \alpha_3, \dots, \alpha_n] \\ &= \cdots = (-1)^{n-1}f'(\alpha_n)[\alpha_1, \alpha_2, \dots, \alpha_{n-1}], \end{aligned} \quad (6)$$

where on the right-hand side each bracketed expression contains one element fewer than the bracketed expression on the left.

Now pose the problem of determining an integral rational function $\varphi(x)$ of degree $(n-1)$ which, for the argument values $x = \alpha_1, \alpha_2, \dots, \alpha_n$, assumes in order the prescribed values $\varphi(\alpha_1), \varphi(\alpha_2), \dots, \varphi(\alpha_n)$.

If one puts

$$\varphi(x) = a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n, \quad (7)$$

then the n unknown coefficients $a_1, a_2, \dots, a_n$ are to be determined from the linear equations

$$\begin{aligned} \varphi(\alpha_1) &= a_1\alpha_1^{n-1} + a_2\alpha_1^{n-2} + \cdots + a_{n-1}\alpha_1 + a_n, \\ \varphi(\alpha_2) &= a_1\alpha_2^{n-1} + a_2\alpha_2^{n-2} + \cdots + a_{n-1}\alpha_2 + a_n, \\ &\vdots \\ \varphi(\alpha_n) &= a_1\alpha_n^{n-1} + a_2\alpha_n^{n-2} + \cdots + a_{n-1}\alpha_n + a_n. \end{aligned} \quad (8)$$

The determinant of this system is precisely the quantity (2), and the problem is therefore always soluble by § 26 when this quantity is different from zero, that is, when no two of the n quantities $\alpha_1, \alpha_2, \dots, \alpha_n$ are equal.

Instead of solving the equations (8) by § 26 and substituting the expressions so found into (7), it is preferable, by § 24, II, to eliminate the homogeneous quantities $1, a_1, a_2, \dots, a_n$ from the $n+1$ equations (7) and (8). This gives the determinant equation

$$\begin{vmatrix} \varphi(x) & x^{n-1} & x^{n-2} & \cdots & x & 1 \\ \varphi(\alpha_1) & \alpha_1^{n-1} & \alpha_1^{n-2} & \cdots & \alpha_1 & 1 \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \varphi(\alpha_n) & \alpha_n^{n-1} & \alpha_n^{n-2} & \cdots & \alpha_n & 1 \end{vmatrix} = 0. \quad (9)$$

We now expand this determinant according to the elements of the first column. For the subdeterminants which occur there we can use the notation (2), and we obtain

$$\begin{aligned} &\varphi(x)[\alpha_1, \alpha_2, \dots, \alpha_n] - \varphi(\alpha_1)[x, \alpha_2, \alpha_3, \dots, \alpha_n] + \varphi(\alpha_2)[x, \alpha_1, \alpha_3, \dots, \alpha_n] \\ &+ \cdots + (-1)^n \varphi(\alpha_n)[x, \alpha_1, \alpha_2, \dots, \alpha_{n-1}] = 0. \end{aligned} \quad (10)$$

Now by (1)

$$[x, \alpha_2, \alpha_3, \dots, \alpha_n] = (x - \alpha_2)(x - \alpha_3) \cdots (x - \alpha_n)[\alpha_2, \alpha_3, \dots, \alpha_n].$$

Dividing (10) by $[\alpha_1, \alpha_2, \dots, \alpha_n]$ and taking account of the several expressions (6), one finally obtains

$$\varphi(x) = f(x) \left\{ \frac{\varphi(\alpha_1)}{(x - \alpha_1)f'(\alpha_1)} + \frac{\varphi(\alpha_2)}{(x - \alpha_2)f'(\alpha_2)} + \cdots + \frac{\varphi(\alpha_n)}{(x - \alpha_n)f'(\alpha_n)} \right\}. \quad (11)$$

This completely determines the function $\varphi(x)$. This expression for $\varphi(x)$ is called the interpolation formula of Lagrange. That it satisfies the stipulated requirements can be verified afterwards very easily by putting $x = \alpha_1, \alpha_2, \dots, \alpha_n$. The denominators $x - \alpha_1, \dots, x - \alpha_n$ are present in it only apparently.